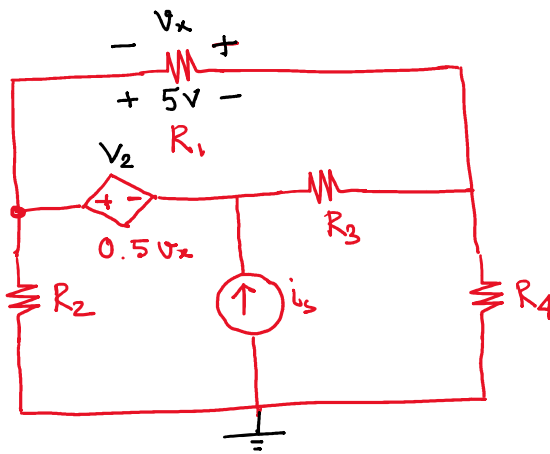
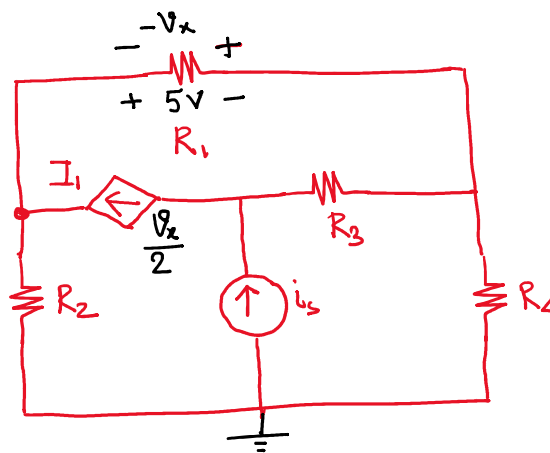




Find V_2 ,

$$\begin{aligned} V_2 &= 0.5V_x, & \text{Now } V_x &= -5V \\ &= 0.5 \times (-5V) \\ &= -2.5V \end{aligned}$$

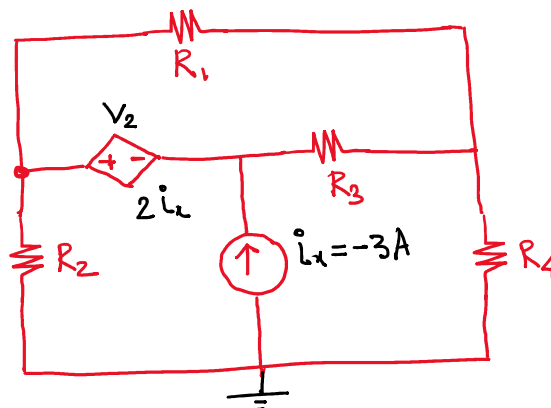
Voltage controlled voltage source



Find I_1 :

$$\begin{aligned} I_1 &= \frac{V_x}{2}, & -V_x &= -5V \\ & & \Rightarrow V_x &= 5V \\ &= \frac{5}{2} A \\ &= 2.5A \end{aligned}$$

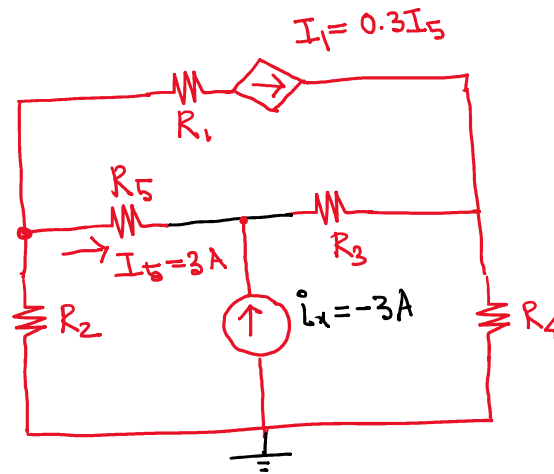
Voltage controlled current source



Find V_2 :

$$\begin{aligned} V_2 &= 2i_x = 2 \times (-3)V \\ &= -6V \end{aligned}$$

Current controlled voltage source

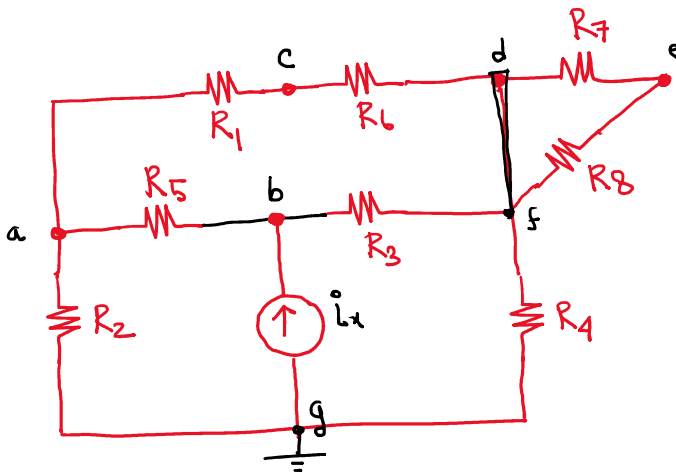


Find I_1

$$I_1 = 0.3I_5$$

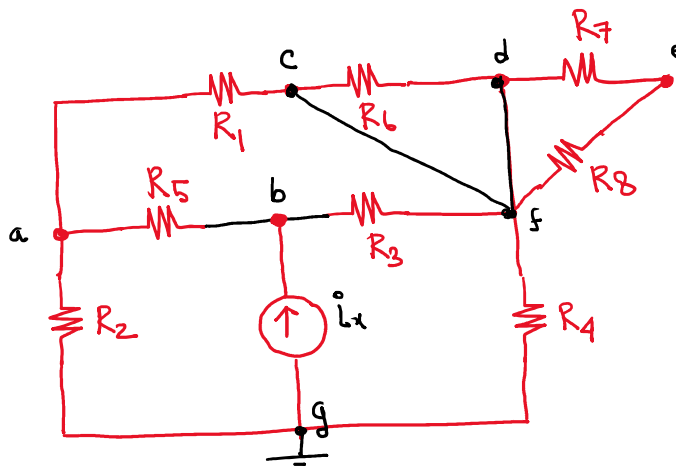
$$= 0.3 \times 3 = 0.9 \text{ A}$$

current controlled current source

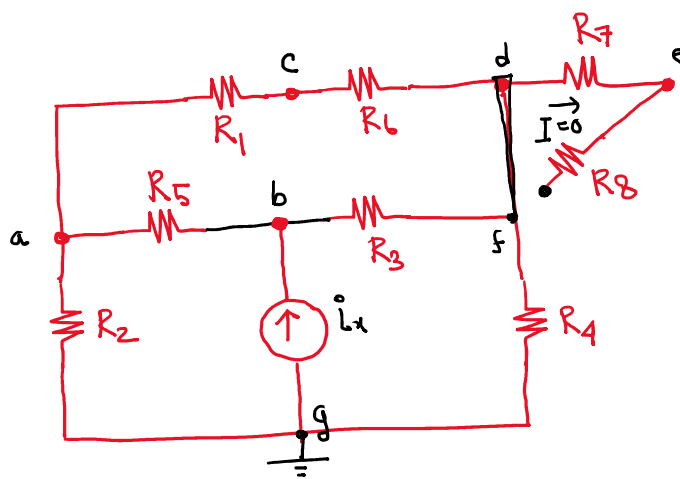


Valid nodes: a, b, c, d, e, f, g
d & f are same node

[No element between d & f]

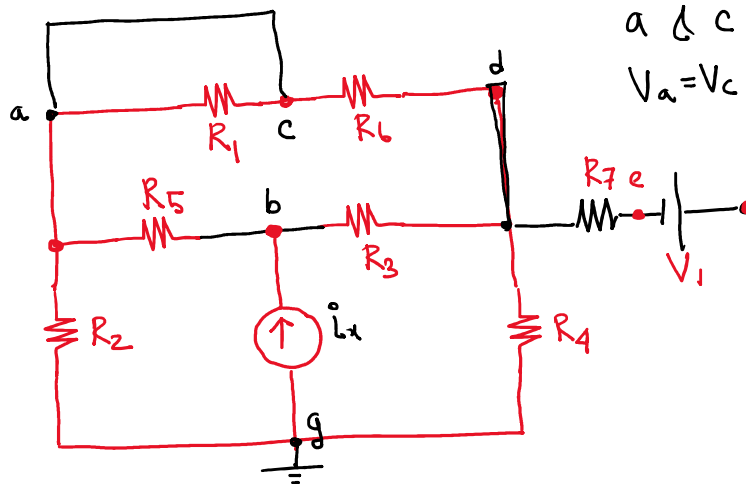


c, f & d are same node



$\square$ d & f Same points,
Same node

$\square$ e is not a valid node.
as it doesn't belong to a
close network (ckt).
floating node.



a & c are same node
 $V_a = V_c$,

e : floating node